Team Members: Ethan Nguyen, Varesh Patel, Najeeb Quadri

Edge AI: Running Quantized LLMs in the Browser (WebGPU)

1. Problem Statement:

Large language models (LLMs) have become increasingly popular in recent years. While they are incredibly powerful, relying on traditional cloud-based APIs (like OpenAI) introduces significant costs, network dependency, and data privacy concerns. "Edge AI"—running models locally on client devices—solves these issues but introduces new constraints regarding hardware capabilities and compute latency.

Our project seeks to determine if client-side LLM inference within a web browser is a viable alternative to cloud APIs for standard consumer hardware.

2. Systems:

We will build a comparative system to measure the performance between Cloud AI and Edge AI.

Edge AI: For this approach, we will run a 4-bit quantized model (e.g., Llama-3-8B or TinyLlama) directly on the main browser thread using the WebGPU API. Doing it this way strips away web development abstractions, giving us access to the device's underlying compute metrics.

Cloud AI (ChatGPT): We will build a secondary, standard API call to a remote model (like gpt-4o-mini) to serve as our control group for network latency versus local compute latency.

3. Evaluation Metrics:

We will pass in the same query for both systems and measure these backend metrics:

+ Time to First Token (TTFT): The latency (in milliseconds) from the moment the prompt is submitted to the moment the first character is generated. This metric is important because it measures the network overhead (for Cloud) vs. model loading/processing time (for Edge).

+ Tokens Per Second (TPS): The rate of text generation once the response begins. This serves as a direct proxy for the raw compute throughput of the underlying hardware.
+ Memory Footprint: The peak RAM and VRAM utilization of the browser tab during the inference cycle.
+ **Performance-to-Quality Ratio:** Since 4-bit quantization inherently degrades model reasoning, we will develop a custom composite metric that weighs generation efficiency (TPS) against output quality. Quality will be evaluated by comparing the Edge AI's response against the Cloud AI's control baseline (e.g., measuring semantic similarity, factual accuracy, or coherence).

<u>4. Experimental Setup:</u>

To evaluate feasibility, we will run this experiment on three different client hardware to understand the hardware variance:

+ High-Performance Node: A gaming laptop equipped with a discrete GPU
+ Unified Memory Node:
    - Macbook Pro M2 max with 30-core GPU and 32GB of RAM
    - Macbook Air with M4 chip, 10-core GPU, and 24GB of RAM.

<u>5. Expected Outcomes:</u>

We aim to find the point where local hardware acceleration becomes a viable substitute for cloud dependency in LLM usage. Our final report will provide a matrix of latency and throughput data across different devices. **Additionally, we will deliver our custom Performance-to-Quality metric to help future developers assess the trade-offs of using quantized models.** Ultimately, this will offer further insights into the possible transition toward running inference in the browser to save costs and preserve client privacy.